
Table of Contents

..... 1

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function ac = aircraft()
% AIRCRAFT Creates an aircraft struct containing default parameters.
%   ac = AIRCRAFT() returns the aircraft struct.

% All units are metric (m-kg-s)

% Initial sizing parameters
% Aircraft parameters
ac.initial.num_crew = 1;
ac.initial.num_eng = 2;
ac.initial.num_drop_tanks = 2;
% Altitudes
ac.initial.h_cruise = 40000 .* 0.3048;           % Cruise altitude (ft to
m)
ac.initial.h_ceiling = 50000 .* 0.3048;          % Ceiling altitude (ft to
m)
ac.initial.h_land = 4000 .* 0.3048;              % Highest elevation
airfoield
% Velocities and Mach numbers
ac.initial.M_cruise = 0.84;                      % Cruise Mach number
ac.initial.V_cruise = getV(ac.initial.h_cruise, ... % Cruise velocity (m/s)
ac.initial.M_cruise);
ac.initial.M_loiter = 0.35;                      % Loiter Mach number
ac.initial.V_climb = 151.2 .* 0.514444;          % Climb speed (kts to m/s)
ac.initial.climb_rate = 55.7 .* 0.514444;        % Rate of climb (kts to
m/s)
ac.initial.climb_angle = deg2rad(13);            % Climb angle (deg to rad)
% Aircraft geometry
ac.initial.b = 60 .* 0.3048;                     % Wing span (ft to m)
ac.initial.Sref = 670 .* 0.092903;               % Planform area (ft^2 to
m^2)
% ac.initial.AR = 4;                             % Aspect ratio
ac.initial.AR = ac.initial.b.^2 ./ ac.initial.Sref; % Aspect ratio
% Aircraft aerodynamics
ac.polar.a2a.clean = simple_polar("clean", ac.initial.num_drop_tanks);
ac.polar.a2a.half = simple_polar("half_flaps", ac.initial.num_drop_tanks);
ac.polar.a2a.full = simple_polar("full_flaps", ac.initial.num_drop_tanks);
ac.polar.a2a.half_gear = simple_polar("half_flaps_gear",
ac.initial.num_drop_tanks);
ac.polar.a2a.full_gear = simple_polar("full_flaps_gear",
ac.initial.num_drop_tanks);
ac.polar.strike.clean = simple_polar("clean", ac.initial.num_drop_tanks);
ac.polar.strike.half = simple_polar("half_flaps", ac.initial.num_drop_tanks);
ac.polar.strike.full = simple_polar("full_flaps", ac.initial.num_drop_tanks);
ac.polar.strike.half_gear = simple_polar("half_flaps_gear",
ac.initial.num_drop_tanks);
ac.polar.strike.full_gear = simple_polar("full_flaps_gear",
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ac.initial.num_drop_tanks);
% Engine performance
ac.initial.TSFC_dry = 0.68 ./ 3600;           % Dry thrust specific
fuel consumption (lb/lb-s to N/N-s)
ac.initial.TSFC_wet = 1.90 ./ 3600;
% ac.initial.TSFC_dry = 0.67 ./ 3600;         % Dry thrust specific
fuel consumption (lb/lb-s to N/N-s)
% ac.initial.TSFC_wet = 1.85 ./ 3600;
ac.initial.T_mil = ac.initial.num_eng .* ...   % Military thrust (lb to
N)
17669 .* 4.44822;
ac.initial.T_max = ac.initial.num_eng .* ...   % Maximum thrust (lb to N)
33093 .* 4.44822;
% ac.initial.T_mil = ac.initial.num_eng .* ... % Military thrust (lb
to N)
% 17595 .* 4.44822;
% ac.initial.T_max = ac.initial.num_eng .* ... % Maximum thrust (lb to
N)
% 29474 .* 4.44822;
% Weights are specified for each mission
% % Aircraft weights
% ac.initial.W0 = 60000 .* 4.44822;           % Takeoff weight (lb to
N)
% ac.initial.We = 30000 .* 4.44822;           % Empty weight (lb to N)
% ac.initial.Wf = 24800 .* 4.44822;           % Fuel weight (lb to N)
% ac.initial.Wf_ext = 10000 .* 4.44822;        % External Fuel
weight (lb to N)
% ac.initial.W_crew = ac.initial.num_crew .* ... % Crew weight (lb to N)
% 200 .* 4.44822;
% ac.initial.W_pay = 2460 .* 4.44822;          % Payload weight (lb to
N)

% Air-to-air mission parameters
ac.a2a.R = 900 .* 1852;                       % Combat radius (nm to m)
ac.a2a.M_dash = 1.7;                          % Dash Mach number
ac.a2a.t_combat = 2.5 .* 60;                  % Combat time (min to s)
ac.a2a.t_loiter = 20 .* 60;                   % Loiter time (min to s)
ac.a2a.h_combat = 10000 .* 0.3048;            % Combat altitude (ft to
m)
ac.a2a.h_dash = 30000 .* 0.3048;             % Combat altitude (ft to
m)
ac.a2a.num_120 = 6;                          % Number of AIM-120
missiles
ac.a2a.num_9x = 2;                          % Number of AIM-9x
missiles
ac.a2a.turn_rate = deg2rad(10);              % Turn rate (deg/s to
rad/s)
ac.a2a.max_g = 8;                            % Maximum vertical load
factor
ac.a2a.max_g_V = getV(ac.a2a.h_combat, 0.95); % Velocity for max load
factor (m/s)
% Weights
ac.a2a.W0 = 60000 .* 4.44822;                % Takeoff weight (lb to N)
ac.a2a.We = 30000 .* 4.44822;                % Empty weight (lb to N)

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ac.a2a.Wf = 24800 .* 4.44822; % Fuel weight (lb to N)
ac.a2a.Wf_ext = 10000 .* 4.44822; % External Fuel weight
(lb to N)
ac.a2a.W_crew = ac.initial.num_crew .* ... % Crew weight (lb to N)
                200 .* 4.44822;
ac.a2a.W_pay = 2460 .* 4.44822; % Payload weight (lb to N)
% Mission segment weight fractions
ac.a2a.Wfracs = [];
ac.a2a.segments = [];

% Strike mission parameters
ac.strike.R = 1020 .* 1852; % Combat radius (nm to m)
ac.strike.M_dash = 0.90; % Combat dash Mach number
ac.strike.V_dash = getV(0, ac.strike.M_dash); % Combat dash velocity
(m/s)
ac.strike.R_combat = 100 .* 1852; % Combat dash distance
(nm to m)
ac.strike.t_combat = ac.strike.R_combat ./ ... % Combat dash time (s)
                    ac.strike.V_dash;
ac.strike.t_loiter = 20 .* 60; % Loiter time (min to s)
ac.strike.h_combat = 0; % Combat altitude (ft to
m)
ac.strike.num_JDAM = 4; % Number of MK-83 JDAMs
ac.strike.num_9x = 2; % Number of AIM-9x
missiles
ac.strike.max_g = 9; % Maximum vertical load
factor
ac.strike.max_g_V = getV(ac.strike.h_combat, 0.95); % Velocity for max load
factor (m/s)
% Weights
ac.strike.W0 = 60000 .* 4.44822; % Takeoff weight (lb to N)
ac.strike.We = 30000 .* 4.44822; % Empty weight (lb to N)
ac.strike.Wf = 24800 .* 4.44822; % Fuel weight (lb to N)
ac.strike.Wf_ext = 10000 .* 4.44822; % External Fuel weight
(lb to N)
ac.strike.W_crew = ac.initial.num_crew .* ... % Crew weight (lb to N)
                200 .* 4.44822;
ac.strike.W_pay = 4424 .* 4.44822; % Payload weight (lb to N)
% Mission segment weight fractions
ac.strike.Wfracs = [];
ac.strike.segments = [];

% Point constraints
ac.pt.seroc_to = 200 .* 0.00508; % Approach SEROC (ft/min
to m/s)
ac.pt.seroc_ap = 500 .* 0.00508; % Takeoff SEROC (ft/min
to m/s)
ac.pt.seroc_to_V = 156 .* 0.514444; % Approach SEROC velocity
(kts to m/s)
ac.pt.seroc_ap_V = 144 .* 0.514444; % Takeoff SEROC
velocity (kts to m/s)

end

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ans =

struct with fields:

initial: [1×1 struct]
polar: [1×1 struct]
a2a: [1×1 struct]
strike: [1×1 struct]
pt: [1×1 struct]

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